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Effect of Pranic Energisation Technique (PET) on psychological well-being of adolescents

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Abstract- Adolescence marks a critical period of transition from childhood to adulthood, characterized by profound physical, emotional, cognitive, and psychological changes. This phase spans around between the ages of 10 and 19. The World Health Organization (WHO) identifies adolescence as a period of particular risk for mental health disorders, often going undetected and untreated. During adolescence, individuals confront numerous challenges including identity formation, peer pressure, academic stress, and hormonal changes, all of which significantly impact their psychological well-being. As per a study by WHO 10-20% globally (WHO) are suffering from Depression, around 6 % globally are in anxiety, and 10% are suffering from substance abuse, Low Self-Esteem, Peer Pressure, Identity Confusion, Bullying (including cyberbullying). To see PET's effect on adolescent psychological well-being. 150 students from an English medium school in Bengaluru, India, between the age of male -13.46±0.34, female-13.49±0.37, were selected. 22 students dropped out from the study due to illness and lack of attendance. PET (n=63), Height-148.57±4.55 male, female144.84±5.22, Weight- male-40.15±3.07 kg, female 40±3.38kg Participants were randomized into two groups, (n=75) participants in the Experimental group practiced PET 4 days a week for 2 months and 2 days a week for the next 2 months as a follow-up, whereas the control group (n=75) did regular activities: total 62 male and 65 female participated DASS-21, ERQ, WISC-IV DLST, Coding, Digit span forward, and Digit span Backward were repeated at the baseline, after 8 weeks, and after 16 weeks again. Statistical Analysis: The descriptive statistics mean; median and standard deviation were obtained. The data was subjected to the Shapiro-Wilks test for normality. The results show that the data does not follow normal distribution (i.e., $p<0.05$). Further Friedman's Test was conducted to see the impact of Pre, Post, and Follow Up. Later Wilcoxon Signed Rank Test was performed to see the Within-group difference. The significant values were compared with 0.05 or 0.01 level of significance. The whole statistical analysis was done using SPSS20. 127 students completed the study.

Keywords: Pranic Energisation Technique, Digit Letter Substitution Test, Digit Backward Test, Emotional Regulation Questionnaire

INTRODUCTION

According to the World Health Organization (WHO), investing in a safe, healthy, and productive transition from childhood to adulthood is vital. Adolescents, numbering 1.3 billion globally, represent 16% of the world's population (Adolescent Data Portal). One in seven individuals aged 10-19 experiences a mental disorder, accounting for 15% of the global disease burden in this age group. Depression, anxiety, and behavioural disorders are among the leading causes of illness and disability in adolescents. Untreated mental health conditions can lead to long-term physical and psychological consequences, hindering the potential for a fulfilling adult life. Adolescence is crucial periods marked by rapid brain development and the acquisition of

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cognitive and socio-emotional skills. Adolescence is a period of transformation in humans with changes in the neural physiology at subcortical and cortical levels. Environmental factors such as violence, parental mental illness, bullying, and poverty significantly increase the risk of mental health issues.[1] Adolescence is a transformative phase and is associated with continued brain maturation and white matter development.[2] Complementary therapies like yoga are increasingly used to support adolescent mental health. In the USA, yoga is among the top five complementary and alternative medicines, practiced by 12% of children. Yoga fosters physical vitality, mental serenity, spiritual growth, and social harmony.

The Pranic Energisation Technique (PET), a cost-effective advance meditation practice developed, needs any initiation.[3] It is safe for all age groups, developed by Dr. H.R. Nagendra at SVYASA University, Bangalore, comprises eight steps and includes hasta mudras to enhance prana.[4] Rooted in Indian philosophy, PET aligns with the Sankhya view that manas, buddhi, chitta, ahankara, and the self (Mahaprana) are subtle forms of prana. WHO further reports that 1.2 billion adolescents face serious mental and physical challenges. Although 15% suffer from mental disorders, most do not seek or receive care. Ignoring the mental and psychosocial needs of youth can severely compromise their future well-being and potential.

MATERIALS & METHODS

This study is a Cluster Randomized Sampling controlled trial. 8-week intervention, 4 days a week, with an 8-week follow-up was done by twice a week. The sample size is calculated using G-power, based on a previous study done on the Effect of PET on university students.[5] The sample size was calculated at 49 for each group by considering the effect size 0.49, alpha 0.05, Power (1-beta error 0.95). Participants were from an English-medium school in North Bangalore, Karnataka.

Selection Criteria- Apparently, healthy subjects were selected, aged 13 to 14 years. Subjects with no prior experience with PET, both boys and girls, were selected. Subjects with asthma, wheezing, learning deficiencies, critical illness, Long-term medication, performing intense sports or physical activity of any type were excluded. Students, who were regular with more than 75% of attendance were kept in the study. Regular practitioners of yoga were also excluded.

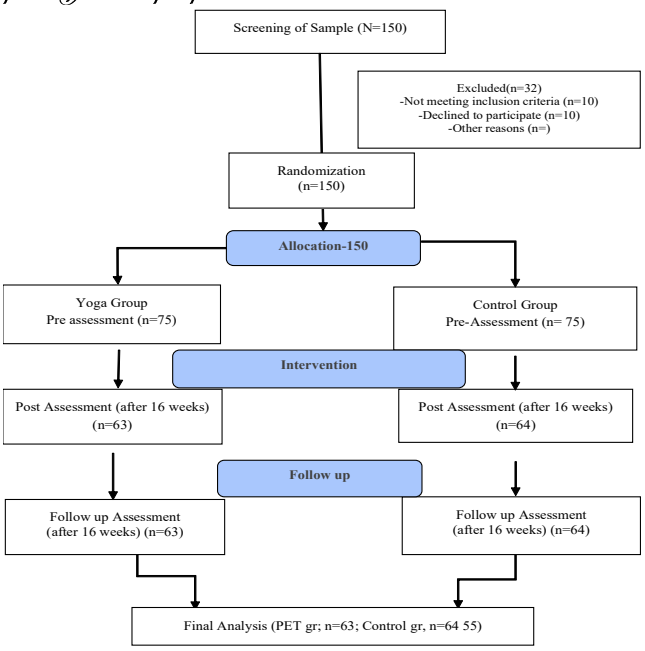


Fig. 1- Consort Flow Chart of the Study

STUDY DESIGN:- This study is a Cluster Randomized Sampling design with 8 weeks of intervention next 8 weeks of follow-up, investigating the effect of the pranic energization technique - An advanced meditation on the psychological well-being of adolescents. After screening them as per the selection criteria participants, the project coordinator assigned students to either the PET or control group (n = 75). The PET group practiced for 8 weeks (30 minutes per day and 4 days per week), 8 weeks follow up, 2 days per week. The control group had regular day-to-day activities without any meditation for the same duration.

INTERVENTIONS:- PET was given (4 days a week; for 30 minutes session) to the intervention group for two months with two months follow-up; twice a week for the next two months; whereas the control group followed their routine work and monitored for the same. The assessors and statisticians were unaware of the intervention assignment status of the subjects. The institutional ethical clearances from SVYASA University, Bengaluru were obtained before starting the trial. Approved by the institutional Ethics committee of SVYASA, the IEC clearance certificate number is RAS/IEC-SVYASA/272/2022. The trial was registered in the Clinical Trial Registry, India (CTRI/2023/06/054410)

ASSESMENT:- All the assessments were done as per the standard guidelines. The evaluations of the Coding, DLST, Digit span forward, Digit span backward, ERQ,

and DASS-21 Scale were performed at baseline, after 8 weeks, and after 16 weeks for both groups.

PSYCHO-SOCIAL ASSESSMENT TOOLS:-

DASS-21(Depression, Anxiety &Stress Scale) (P. Lovibond & S. Lovibond (1995) (Questions-21) The DASS-21 is a well-established instrument for measuring depression, anxiety, and stress with good reliability and validity, developed by Lovibond and Lovibond (1995). The DASS-21, with its 21 items (7 items for each subscale) and three dimensions with similar psychometric properties, is based on the tripartite model of depression, anxiety, and stress.

Emotional Regulation Questionnaire (ERQ)-

Developed by Gross & John (2003)[6], It is meant for 8-18 years old Students, it is a 10-item Likert scale, this is a psychometric evaluation of the Emotion Regulation Questionnaire for Children and Adolescents (ERQ-C) It investigates the use of 2 specific strategies of that is suppression and reappraisal.[6] Respondents' answers are scored on a 7-point Likert-type scale ranging from 1 (strongly disagree) to 7 (strongly agree). The scoring takes the average of all the scores in each subscale of cognitive reappraisal and expressive suppression. The higher the score, the greater the use of that particular emotion regulation strategy, conversely lower scores represent less frequent use. It was administered individually.

COGNITIVE TOOLS

The Wechsler Intelligence Scale for Children IV (WISC): - Developed by David Wechsler (2003) is an individually administered intelligence scale for 6 to 16-year-old children, it represents a child's general intellectual ability.

Digit-Symbol Coding-(WISC): - Coding subtest (CD) of the Wechsler Adult Intelligence Scale-Fourth Edition (WAIS,2008) primarily measures processing speed and clerical efficiency. This subtest is complex and require attention, concentration, cognitive flexibility, visual scanning, adequate motor coordination, and positive test taking motivation[7] for 6 -16-year-old children, R=.85, Each of a series of shapes or numbers is paired with a simple symbol, based on a key; the examinee draws the symbol under the corresponding number. Participants have to look carefully at the key that matches numbers with symbols. Then write the number that goes with each symbol in the space below it. This was administered in the group, the time limit was 90 seconds, and participants could code horizontally or vertically as they wished.[7]

Digit Backward and Digit Forward Test- Is designed by David Wechsler, for the age group of 6 -16 years, (reliability score is =.85). There are two parts to this subtest. Digits forward and digits backward. The test was developed to measure verbal short-term memory and attention. In both the tests were administered individually, the researcher read out the digit to the individual participant and in the forward digit, they repeated the numbers in the same order only once, for Digit span backward: -The digits were read for the individual participants and they repeated them in backward order. The scoring was done for both DIGIT SPAN forward and backward separately. one of the best normed and most highly respected measures of child intelligence. Each tap has distinct but interdependent cognitive functions. Digits Forward primarily taps short-term auditory memory while Digits Backward measures the child's ability to manipulate verbal information while in temporary storage.

Digit Letter Substitution Test: - The DLST, a subtest of WISC IV, consists of an 8×12 array of random digits (1-9). Subjects start with the worksheet upside down and are given a coding sheet to substitute specific letters for each digit. They choose their substitution strategy (horizontal, vertical, or random) and have 90 seconds to substitute as many digits as possible. The test is timed with a standard stopwatch.[8] Administered in a group, subjects turn over the worksheet and stop after 90 seconds. Marks are later calculated by the researcher.

Data Extraction and Analysis: - Data was collected from Subjects individually on day 1, after 8 weeks, and after 16 weeks for both the PET and the control group. Data entry and extraction were done under the guidance of a statistician. All forms/questionnaires/tools were thoroughly screened.

Statistical Analysis: The descriptive statistics mean, median, and standard deviation were obtained.

- The data was subjected to the Shapiro-Wilks test for normality.
- The results show that the data does not follow normal distribution (i.e., $p<0.05$).
- Friedman's Test was conducted to see the impact of Pre, Post and Follow Up.
- Later the Wilcoxon Signed Rank Test was performed to see the within group difference.
- The significant values were compared with 0.05 or 0.01 levels of significance.
- The whole statistical analysis was done using SPSS 20.

PSYCHOLOGICAL VARIABLE

Table 1: Within group Comparisons of DASS and ERQ scores

Variables	Groups	MEAN±SD			Pre vs post			Pre vs Follow up	
		Pre (Baseline)	Post (8weeks)	Follow up (16weeks)	% Change	P value	% Change	P Value	Z Score
Depression	PET	13.17±8.92	6.0±7.40	6.89±6.79	-54%	0.000**	47.60%	0.000**	3.68
	CON	20.44±7.57	18.72±6.31	20.50±7.65	-8.4%	0.16	0.29%	0.163	3.63
Anxiety	PET	13.21±8.63	9.56±5.951	8.10±5.55	-27.6%	0.009**	38.68%	0.000**	7.83
	CON	20.50±8.85	20.97±7.71	19.94±6.66	2.29%	0.000**	2.73%	0.000**	7.09
Stress	PET	13.08±7.15	8.67±4.79	8.35±5.07	-33.70	0.01*	-36.10%	0.000**	3.33
	CON	18±9.58	20.75±10.95	22.50±8.98	15.27%	0.000**	25%	0.000**	7.75
ERQ	PET	4.45±0.89	4.48±0.08	4.43±0.77	3.82%	0.237	0.44%	0.164	1.18
	CON	4.35±0.83	4.31±0.93	3.99±0.82	0.91%	0.968	8.27%	1.392	0.164
Reappraisal	PET	4.39±1.11	4.32±1.06	4.25±1.15	-1.5%	0.88	-3.1%	0.127	1.53
	CON	4.62±1.08	4.56±0.95	4.49±1.02	-1.2%	0.380	-2.81%	0.23	1.201
Suppre	PET	4.19±1.04	4.35±1.12	4.33±1.01	3.8%	0.500	3.3%	0.666	0.5
	CON	4.29±1.23	4.30±1.24	3.60±0.98	0.23%	0.000**	-16.08%	0.042*	3.951

Table 2: Between-group comparisons DASS and ERQ Scores

Variables	Groups	Pre (Baseline)	Post (8weeks)		Follow up (16weeks)		
		Mean±SD	P Value	Mean±SD	P value	Mean±SD	P Value
Depression	PET	13.17±8.92	0.000**	6.0±7.40	0.000**	6.89±6.79	0.000**
	Control	20.44±7.57		18.72±6.31		20.50±7.65	
Anxiety	PET	13.21±8.63	0.000**	9.56±5.951	0.000**	8.10±5.55	0.000**
	Control	20.50±8.85		20.97±7.71		19.94±6.66	
Stress	PET	13.08±7.15	0.000**	8.67±4.79	0.000**	8.35±5.07	0.000**
	Con	18±9.58		20.75±10.95		22.50±8.98	
ERQ	PET-	4.45±0.89	0.46	4.48±0.08	0.000**	4.43±0.77	0.006
	CON-	4.35±0.83		4.31±0.93		3.99±0.82	
Reapp	PET	4.39±1.11	0.237	4.32±1.06	0.000**	4.25±1.15	0.23
	CON	4.62±1.08		4.56±0.95		4.49±1.02	
Suppr	PET	4.19±1.04	0.505	4.35±1.12	0.000**	4.33±1.01	0.000**
	CON	4.29±1.23		4.30±1.24		3.60±0.98	

RESULT

DASS: Decrease in depression from baseline to post by 54% and a further 47.60 % decrease is seen i.e., from 20.44% to 18.72% and later an increase to Follow up i.e., to 20.50%. It is statistically significant ($p<0.01$) for the PET group and insignificant for the Control group the change is 8.4% and a further 0.29% ($p>0.05$), From Mann Whitney U Test and Wilcoxon Signed Rank Test. Anxiety reduced significantly-27.6% ($p<0.05$) by 8th week and by 16th week it further reduced by-38.6% for the PET group and statistically insignificant change for the Control group ($p>0.05$) 2.29% decrease by from 8th week and again increased 2.73% by 16th week. Stress reduced by -33.70% in 8 weeks and further decreased by -36.10% by 16th week In PET group Whereas in the Control group stress increased by 15.20% by 8th week and later 25% in Follow up phase. It is statistically significant ($p<0.05$) for the PET group and statistically insignificant for the Control group ($p>0.05$). Pairwise difference on Anxiety scores for PET Group: Wilcoxon Signed Rank Test, after 8th week p was 0.009** and after 16th week there is significant change of 0.000**.

ERQ: -In the PET group, the emotional regulation strategy decreased from 0.67% in 8 weeks then slightly decreased by -0.44% by the 16th week. In the Control group, the values came down from -0.91% in 8 weeks then by the 16th week -8.27% change. From Friedman's Test, it is seen that change in control is statistically

insignificant (p<0.01) while the PET group is statistically significant (p>0.05). Reappraisal in all cases was observed to be statistically non-significant (p>0.05). -1.5% low in the PET group by 8 weeks, then -3.1% by week which is insignificant. By Mann Whitney U Test, it is observed that after 16th week changes were significant (p<0.01). From the Friedman Test, it can be seen that it was statistically insignificant (p<0.01) for both groups. The pairwise difference in the Control group- pre to post Z is 0.88, post to follow up- Z value -0.84, for PET group, Reappraisal groupwise is P value- 0.91 for control group- p is 0.04. In the suppression PET group, the mean score ↑d by 3.8% by 8 weeks and by 16 weeks 3.3%change In PET group is statistically significant (p<0.01) while Control group is statistically insignificant (p>0.05). Within group difference on PET Group of Suppression scores: Wilcoxon Signed Rank Test pre to post 2.66% change which is not significant (p<0.01) but post to follow up there 87.20 % change which is significant (p>0.05).

Table 3: Groupwise effect of PET on DLST Tests: Descriptives and Mann Whitney U Test

Variables	Baseline Mean±SD	8 week	16 week	within group % changes		Z1	P1	Z2	P2	χ^2 (df=2)
				pre to post	pre to f. up					
DFT PET	6.43±2.11	8.03±1.60	8.30±1.72	24.88%	29.08%	5.02	0.000**	5.05	0.000**	43.36
CON	5.95±2.05	5.13±1.58	4.19±1.54	-13.78	-29.57	4.7	0.000**	6.1	0.000**	64.87
Coding PET	45.51±7.35	50.68±8.96	55.05±40.31	11.36%	20.96%	4.04	0.001**	3.44	0.000**	39.38
Coding CON	34.22±8.52	34.58±7.77	40.31±12.21	1.05%	17.76%	1.39	0.165	3.38	.001**	8.24
DBT PET	4.02±1.99	5.46±2.05	5.05±2.29	35.8%	25.62%	4.68	0.001*	2.60	0.009	19.26
DBT CON	2.84±1.87	2.30±1.52	1.53±1.35	-19.01%	-46.12%	4.9	0.000**	6.32	0.000**	82.7
DLST PET	48.57±11.04	52.13±10.91	58.03±9.45	19.68%	19.47%	1.75	0.08	4.93	0.000**	28.94
DLST CON	42.55±11.75	36.36±10.36	45.09±11.75	-14.54	5.96%	3.46	0.001*	1.52	0.128	19.44

COGNITIVE ANALYSIS

Results of cognitive variables: -for the Digit Forward Test primarily taps short-term auditory memory analysis, By Mann Whitney U Test we saw 24.88% improvement by the 8th week and 29.08% change noticed in the PET group and by 16 weeks, which is statistically significant (p<0.05) in all cases. The Coding questionnaire tests short-term memory and concentration. In Coding by Friedman's Test, In the PET group, pre- to post-phase, 11.36% improvement, after 16 week-20.96% improvement was seen which is significant. In the Control group an ↑ of 1.05% after 8 weeks, and 17.76% change was observed after 16 weeks.in the Control group, Post to Follow up and Pre to Follow up were seen to be statistically significant (p<0.01). However, Pre to Post in the Control group is insignificant (p>0.05). Pairwise p value is 0.00** for the PET group, p value is 0.00**, Control group- Z value by 8th week P value is 0.16, by 16th week P value is 0.000**.

Digit Backward Test: measures the child's ability to manipulate verbal information while in temporary memory. In PET group, by 8th week ↑ of 35.8% and ↓s to 25.62% decreased by 16th week which is statistically significant (p<0.01). However, in the Control group a decrease was seen from Pre to post -19.01% and later a decrease of -46.12% in the follow-up phase. From Friedman's Test, it can be seen that it is statistically significant (p<0.01) in PET. Whereas, in the Control group statistically insignificant (p<0.01), which helps to prove the hypothesis that PET can help to improve cognition. Within-group difference on DBT OF ON PET Group: Wilcoxon Signed Rank Test, in Pre to post z value is 0.04, and insignificant change in the control group. Whereas the post to Follow Up Z value is 1.85 and a significant change of 0.00**.

Pairwise- pre to follow up Z =1.97 and p value is 0.049*, in control group, Pre to post Z = 5.53, p value is 0.000** significant, pre to follow up is z=6.42, p value is 0.000** also significant. Anxiety scores ↑d from pre-phase

(16.88) to post-phase (15.31) and further to follow-up (14.06). Friedman's Test was statistically insignificant ($p>0.05$). Anxiety Groupwise: PET group: Pre to Post phase showed a decreased (13.21 to 9.56) and a further decrease in Follow-up (8.10). Control group: Pre to Post phase anxiety ↑d (20.50 to 20.97) and then ↓d in Follow-up (19.94). Statistically significant improvement in the PET group ($p<0.05$); the Control group remained insignificant. In summary, PET had a positive impact on depression. However, anxiety scores showed less consistent improvement.

DISCUSSION

This study found that stress, anxiety, depression levels decreased and improvement in short term memory, processing speed in the PET group. This is supported by a systematic review of 16 RCTs, including a Cochrane review, confirmed that meditation benefits children aged 5-18 with ADHD. Daily meditation (10-15 min, twice a day) in schools has shown positive outcomes. In RCTs in India and the U.S., 79 adolescents (13-15 years) practicing Transcendental Meditation (TM) for 1-2 years demonstrated improved attention and conflict resolution compared to 76 controls.[9] TM in six low-income schools over two years also reduced student arrests and increased attendance. A U.S.-based RCT on 11-12-year-olds (n=52) showed that 12 minutes of mindfulness daily for 6 weeks reduced disturbances, suicidal thoughts, and self-harm. Another large RCT (n=201, aged 13-20) reported decreased depression after 8 weeks of weekly mindfulness sessions (100 minutes) and daily home practice (15 minutes), compared to 207 controls. A cohort study on 38 adolescents (16.5 years old) found structural brain improvements in regions related to emotional and physical awareness after a 12-week mindfulness program, compared to 21 non-participants. Sankhla *et al.* (2014)[5] demonstrated immediate effects of yogic techniques like Nadi Shuddhi, Bhramari, OM meditation, PET, and Mind Sound Resonance on brain wave coherence. Controls showed that these practices enhance brain orderliness, vital for academic performance. Super Brain Yoga improved concentration by 70.5% in Mysore school students. Bhramari pranayama was linked to parasympathetic dominance, reduced heart rate and BP, improved cognition, reduced tinnitus-related irritability, EEG changes, and lower stress.[10]

The conceptual basis lies in the Taittiriya Upanishad's Panchkosha model, describing five sheaths: Annamaya, Pranamaya, Manomaya, Vijnanamaya, and Anandamaya kosha. Disharmony between Vijnanamaya and Manomaya leads to mental disturbances that eventually manifest physically. PET, developed by Dr. H.R. Nagendra (2007)[3], is an advanced, cost-effective meditation working on Vyana prana using breath regulation, hasta mudras, awareness rotation, and silence. It helps harmonize the Manomaya and Pranamaya koshas, reducing stress and enhancing vitality. This study explored PET's effects on emotional well-being, depression, anxiety, stress, attention, and working memory. Prior studies show yoga and meditation improve academic performance and behaviour in ADHD students. Different meditation forms influence brain function supported by scientific facts: Loving-kindness/ compassion activates limbic regions.

- Shamata enhances the frontal-parietal attention network.
- Mindfulness deactivates medial but activates lateral prefrontal cortices.
- Vipassana shows bilateral activation in the rostral anterior cingulate and dorsal medial prefrontal cortex.
- Zen increases alpha and theta activity in multiple brain regions, including the frontal cortex.
- Yoga Nidra activates the hippocampus and imagery areas, while deactivating the executive control network.[11]
- Transcendental Meditation is characterized by frontal alpha1 (8-10 Hz) coherence.

CONCLUSION

Both the PET and control groups demonstrated improvements in psychological well-being. However, the PET group showed significantly greater efficacy in enhancing cognitive function and reducing stress, depression, and anxiety. These findings suggest that the Pranic Energisation Technique, which is an advanced meditation, may be a promising, cost-effective intervention for promoting psychological health among adolescents. Further large-scale, longitudinal studies are warranted to validate these outcomes and establish PET as a reliable tool for fostering holistic development, supporting the integration of a sound mind and body in adolescence.

FINANCIAL SUPPORT AND SPONSORSHIP

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CONFLICTS OF INTEREST

There are no conflicts of interest

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